

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1610

COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
Weightage	25%	25%	25%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Tejaswini.M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID Symptoms Observed

P1 Fever, Cough
P2 Fever, Headache
P3 Cough, Headache
P4 Fever, Cough, Headache
P5 Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "**Fever**". Identify the valid association rules and justify why they satisfy the given constraints.

Given:

- Minimum support = **50%**
- Minimum confidence = **70%**
- Every generated rule must include **Fever**.

Dataset

Patient Symptoms

P1 Fever, Cough
P2 Fever, Headache
P3 Cough, Headache
P4 Fever, Cough, Headache
P5 Fever, Fatigue

Total transactions = **5**

Step 1: Calculate support

Fever

Fever occurs in P1, P2, P4, P5.

$$\text{Support}(\text{Fever}) = \frac{4}{5} \times 100 = 80\%$$

So Fever is frequent.

Fever + Cough

Occurs in P1 and P4.

$$\text{Support}(F, \text{everCough}) = \frac{2}{5} \times 100 = 40\%$$

Below 50% → rejected.

Fever + Headache

Occurs in P2 and P4.

$$\text{Support}(F, \text{everHeadache}) = \frac{2}{5} \times 100 = 40\%$$

Below 50% → rejected.

Fever + Fatigue

Occurs only in P5.

$$\text{Support}(F, \text{everFatigue}) = \frac{1}{5} \times 100 = 20\%$$

Rejected.

Step 2: Check possible rules

For:

Fever → Cough

Support = 40%, which is less than 50%.

For:

Cough → Fever

Support = 40%, also less than 50%.

Similarly:

- Fever → Headache = 40% support
- Headache → Fever = 40% support
- Fever → Fatigue = 20% support
- Fatigue → Fever = 20% support

Final Answer

No valid association rule satisfies all the constraints.

Rule	Support	Result
Fever → Cough	40%	Reject
Cough → Fever	40%	Reject
Fever → Headache	40%	Reject
Headache → Fever	40%	Reject
Fever → Fatigue	20%	Reject
Fatigue → Fever	20%	Reject

Justification: Although Fever itself has 80% support, every two-item combination containing Fever has support below the required 50%. Therefore, no association rule can satisfy the minimum-support constraint, regardless of confidence.

Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

Step 1: Generate 2-itemsets

{Python, SQL}

Occurs in:

- S1
- S4

$$Support = \frac{2}{5} = 40\%$$

✓ Frequent

{Python, Data Science}

Occurs in:

- S2
- S4

$$Support = \frac{2}{5} = 40\%$$

✓ Frequent

{Python, Machine Learning}

Occurs only in S5.

$$Support = \frac{1}{5} = 20\%$$

✗ Not frequent

{SQL, Data Science}

Occurs only in S4.

$$Support = \frac{1}{5} = 20\%$$

✗ Not frequent

{SQL, Machine Learning}

Occurs only in S3.

$$Support = \frac{1}{5} = 20\%$$

✗ Not frequent

{Data Science, Machine Learning}

Occurs in no student.

$$Support = 0\%$$

✗ Not frequent

Step 2: Check 3-itemsets

Possible 3-itemsets do not occur in at least 2 students.

Therefore, none satisfy the 40% minimum-support requirement.

Final Answer

The frequent itemsets satisfying both constraints are:

Frequent Itemset	Support
{Python, SQL}	40%
{Python, Data Science}	40%

How does the constraint help pruning?

The **minimum itemset size constraint** eliminates all single-course itemsets such as {Python} and {SQL} because the question requires at least two courses.

It reduces the search space because the algorithm only considers combinations containing **two or more courses**.

Also, the **40% support constraint** eliminates combinations appearing in fewer than 2 students.

Conclusion: The constraints significantly reduce candidate generation and make association-rule mining faster.

Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Step 1: 1-itemsets

Item	Count	Support
Data Mining	4	80%
Python Programming	2	40%
Database Systems	3	60%
Artificial Intelligence	1	20%

Step 2: 2-itemsets

{Data Mining, Python Programming}

T1, T4 → 2 transactions

$$Support = \frac{2}{5} = 40\%$$

{Data Mining, Database Systems}

T2, T4 → 2 transactions

$$Support = \frac{2}{5} = 40\%$$

{Data Mining, Artificial Intelligence}

T5 → 1 transaction

$$Support = 20\%$$

{Python Programming, Database Systems}

T3, T4 → 2 transactions

$$Support = 40\%$$

{Python Programming, Artificial Intelligence}

0 transactions → 0%

{Database Systems, Artificial Intelligence}

0 transactions → 0%

Step 3: 3-itemsets

{Data Mining, Python Programming, Database Systems}

Only T4 contains all three.

$$Support = \frac{1}{5} = 20\%$$

Other possible 3-itemsets do not occur.

Constraint: Maximum size = 3

Therefore, Apriori **stops after generating 3-itemsets**.

There is no need to generate:

4-itemsets, 5-itemsets, ...

Final Answer

The repeated 2-itemsets are:

Itemset	Support
Data Mining, Python Programming	40%
Data Mining, Database Systems	40%
Python Programming, Database Systems	40%

The only 3-itemset is:

Itemset	Support
Data Mining, Python Programming, Database Systems	20%

How does the constraint reduce complexity?

Without the constraint, Apriori could continue generating larger combinations.

With:

$$\text{Maximum Itemset Size} = 3$$

the algorithm stops after 3-itemsets. Therefore, candidate generation and support counting for larger itemsets are completely avoided.

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "**Computer Science**" level. Explain how the course hierarchy affects the discovery of association rules.

Step 1: Focus only on Computer Science level

We ignore rules involving **Engineering** and **Information Technology**, because the constraint says: Rules should be generated only at the **Computer Science** level.

Computer Science occurs in:

- S1
- S3
- S4

Therefore:

$$\text{Support}(\text{Computer Science}) = \frac{3}{5} = 60\%$$

Step 2: Find related courses

Machine Learning occurs together with Computer Science in:

- S1

- S4

Therefore:

$$\text{Support}(C, SML) = \frac{2}{5} = 40\%$$

Rule 1: Computer Science → Machine Learning

$$\begin{aligned}\text{Confidence} &= \frac{\text{Support}(C, SML)}{\text{Support}(CS)} \\ &= \frac{2/5}{3/5} \\ &= 66.67\%\end{aligned}$$

So:

Computer Science → Machine Learning

- Support = **40%**
- Confidence = **66.67%**

Reverse Rule

Machine Learning → Computer Science

Machine Learning occurs in S1 and S4 = 2 students.

$$\text{Confidence} = \frac{2}{2} = 100\%$$

Therefore:

- Support = **40%**
- Confidence = **100%**

Final Answer

Rule	Support	Confidence
Computer Science → Machine Learning	40%	66.67%
Machine Learning → Computer Science	40%	100%

Effect of the hierarchy

The hierarchy allows mining at a specific level instead of considering every course category.

For example:

Engineering → Computer Science → Machine Learning

The constraint tells us to focus specifically on **Computer Science**, reducing irrelevant rules and making the discovered relationships more meaningful.

Q5. Dataset: Insurance Customer Records

CID Age Group Income Level Policy Purchased

C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Step 1: Apply constraint Age Group = Young

Keep only:

Customer	Age	Income	Policy
C1	Young	High	Health Insurance
C3	Young	Low	Travel Insurance
C5	Young	Medium	Vehicle Insurance

Total Young customers = 3

Step 2: Analyze income levels

Among Young customers:

- High income = 1
- Low income = 1
- Medium income = 1

Therefore:

$$\begin{aligned} \text{High} &= 33.33\% \\ \text{Low} &= 33.33\% \\ \text{Medium} &= 33.33\% \end{aligned}$$

Step 3: Analyze policy types

Among Young customers:

- Health Insurance = 1
- Travel Insurance = 1
- Vehicle Insurance = 1

Each represents:

$$\frac{1}{3} \times 100 = 33.33\%$$

Meaningful patterns

Pattern 1

Young + High Income → Health Insurance

Observed for C1.

Pattern 2

Young + Low Income → Travel Insurance

Observed for C3.

Pattern 3

Young + Medium Income → Vehicle Insurance

Observed for C5.

Final Answer

Age Group Income Level Policy

Young	High	Health Insurance
Young	Low	Travel Insurance
Young	Medium	Vehicle Insurance

Marketing strategy

These patterns can help an insurance company create targeted campaigns:

- **Young + High income** → promote premium/health insurance plans.
- **Young + Medium income** → promote affordable vehicle insurance.
- **Young + Low income** → promote low-cost travel insurance.

However, the dataset contains only **three Young customers**, so these patterns should be treated as preliminary rather than strong general-market associations.